United States Patent	12384163
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Hayashi; Masahiro et al.

Tank connectable to liquid container, and liquid supplying system including tank and liquid container

Abstract

A tank includes: a tank body defining a storage chamber; a recessed portion provided at the tank body; a slider positioned in the recessed portion; a fitting portion provided at the slider; and a sealing member. The recessed portion has an injection opening in communication with the storage chamber, and a connection opening through which a liquid container is connectable. The slider includes a nozzle extending toward the connection opening in an extending direction and having a communication opening communicable with the injection opening. The slider is movable toward and away from the connection opening in the extending direction while rotating relative to the recessed portion. The fitting portion can be fitted with the liquid container. The sealing member can provide a liquid tight sealing to a gap between the slider and the recessed portion, and allow liquid flow between the communication opening and the injection opening.

Inventors: **Hayashi; Masahiro (Nishio, JP), Okazaki; Naoya (Hashima, JP), Oki; Satoru (Nagoya, JP), Abe; Nanami (Nagoya, JP)**

Applicant: **BROTHER KOGYO KABUSHIKI KAISHA** (Nagoya, JP)

Family ID: **84490929**

Assignee: **BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)**

Appl. No.: **18/596759**

Filed: **March 06, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240208235 A1	Jun. 27, 2024

Foreign Application Priority Data

JP	2021-101637	Jun. 18, 2021

Related U.S. Application Data

continuation parent-doc US 17834058 20220607 US 11951747 child-doc US 18596759

Publication Classification

Int. Cl.: **B41J2/175** (20060101)

U.S. Cl.:

CPC **B41J2/17523** (20130101);

Field of Classification Search

CPC: B41J (2/17513); B41J (2/17596); B41J (2/175); B41J (2/17506); B41J (2/17523); B41J (2/17509); B41J (2/1754); B41J (2/17553); B41J (2/17566); B41J (29/02); B41J (29/13)

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Primary Examiner: Lebron; Jannelle M

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/834,058 filed on Jun. 7, 2022, now U.S. Pat. No. 11,951,747, which claims priority from Japanese Patent Application No. 2021-101637 filed on Jun. 18, 2021. The entire contents of the aforementioned applications are incorporated herein by reference.

BACKGROUND ART

(1) In a conventional printing device, there has been known a structure for supplying ink to a tank from a container connected to the tank each time the ink stored in the tank is consumed. The container has a supply opening for supplying the ink. In order to prevent the ink leakage through the supply opening, conceivably, a conventional supply valve provided in an ink cartridge may be employed in the container. Further, in order to promote smooth outflow of the ink through the supply opening of the container, preferably, the container may have an air communication passage. In order to open and close the air communication passage, an air releasing valve may be employed in the container.

SUMMARY

(2) The conventional supply valve and the air releasing valve described above are urged by springs to close the supply opening and the air communication passage, respectively. If such valves are provided in the container, a user needs to push the container against the tank or to keep the container pressed against the tank, against the urging force of the springs, when connecting the container to the tank. Further, the container is likely to bound off the tank due to the urging force of the springs when the user attempts to remove the container from the tank.

(3) In view of the foregoing, it is an object of the present disclosure to provide a structure capable of enhancing user's convenience upon attachment and detachment of the container to and from the tank.

(4) In order to attain the object, according to a first aspect, the disclosure provides a tank to which a liquid container is connectable. The tank includes a tank body, a recessed portion provided at the tank body, a slider positioned in the recessed portion, a fitting portion provided at the slider, and a sealing member. The tank body defines a tank storage chamber configured to store liquid therein. The recessed portion has an injection opening in communication with the tank storage chamber and a connection opening through which the liquid container is connectable. The slider is rotatable relative to the recessed portion. The slider includes a nozzle extending toward the connection opening in an extending direction. The nozzle has a communication opening configured to communicate with the injection opening of the recessed portion. The slider is movable toward and away from the connection opening in the extending direction relative to the recessed portion in accordance with rotations of the slider. The fitting portion is configured to be fitted with the liquid container. The sealing member is configured to provide a liquid tight sealing to a gap between the slider and the recessed portion, and configured to also allow the liquid to flow between the communication opening of the nozzle and the injection opening of the recessed portion.

(5) With this structure, as the liquid container fitted with the fitting portion of the slider is rotated, the slider rotates relative to the recessed portion and slidingly moves together with the nozzle. When the nozzle enters into a supply opening of the liquid container, the liquid in the liquid container flows into the tank storage chamber through the nozzle.

(6) According to another aspect, the disclosure provides a liquid supplying system including a liquid container and a tank. The liquid container includes a casing, a container storage chamber, a supply opening, and a counter-fitting portion. The container storage chamber is defined in the casing and is configured to store liquid therein. The supply opening provides fluid communication between the container storage chamber and an outside of the casing. The counter-fitting portion is provided at the casing. The tank includes a tank body, a recessed portion, a slider, a first fitting portion, and a sealing member. The tank body defines therein a tank storage chamber configured to store therein liquid. The recessed portion is provided at the tank body. The recessed portion has an injection opening in communication with the tank storage chamber and a connection opening through which the liquid container is connectable. The slider is positioned in the recessed portion and is rotatable relative to the recessed portion. The slider includes a nozzle extending toward the connection opening in an extending direction. The nozzle has a communication opening configured

to communicate with the injection opening of the recessed portion. The slider is movable toward and away from the connection opening in the extending direction relative to the recessed portion in accordance with rotations of the slider. The first fitting portion is provided at the slider and is configured to be fitted with the counter-fitting portion. The sealing member is configured to provide a liquid tight sealing to a gap between the slider and the recessed portion, and is configured to also allow the liquid to flow between the communication opening of the nozzle and the injection opening of the recessed portion.
(7) With this structure, as the liquid container is rotated with the first fitting portion on the slider fitted with the counter-fitting portion of the liquid container, the slider rotates relative to the recessed portion and slidingly moves together with the nozzle. When the nozzle of the slider enters into the supply opening of the liquid container, the liquid in the container storage chamber flows into the tank storage chamber through the nozzle.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The particular features and advantages of the embodiment(s) as well as other objects will become apparent from the following description taken in connection with the accompanying drawings, in which:
(2) FIG. **1** is a perspective view of a multifunction device **10**;
(3) FIG. **2** is a vertical cross-sectional view schematically illustrating an internal structure of a printer portion **11** of the multifunction device **10**;
(4) FIG. **3** is a perspective view of a tank **70** of the multifunction device **10**;
(5) FIG. **4** is a vertical cross-sectional view of the tank **70** taken along a cross-section containing an axis **77**X of the tank **70** and extending in an up-down direction **7**;
(6) FIG. **5** is a perspective view of a bottle **100**;
(7) FIG. **6** is a cross-sectional view of the bottle **100** taken along a cross-section containing an axis **100**X of the bottle **100** and extending in the up-down direction **7**;
(8) FIG. **7** is a perspective view illustrating the tank **70** and the bottle **100** prior to the insertion of the bottle **100** to a recessed portion **77** of the tank **70**;
(9) FIG. **8** is a cross-sectional view illustrating the tank **70** and the bottle **100** prior to the insertion of the bottle **100** to the recessed portion **77** of the tank **70** taken along the cross-section containing the axis **77**X and extending in the up-down direction **7**;
(10) FIG. **9** is a cross-sectional view illustrating the tank **70** and the bottle **100** inserted in the recessed portion **77** of the tank **70** taken along the cross-section containing the axis **77**X and extending in the up-down direction **7**;
(11) FIG. **10** is a cross-sectional view illustrating the tank **70** and the bottle **100** inserted in the recessed portion **77** of the tank **70** taken along the cross-section containing the axis **77**X and extending in the up-down direction **7**, and particularly illustrating a state where the bottle **100** is rotated counterclockwise from the state illustrated in FIG. **9**;
(12) FIG. **11** is a cross-sectional view illustrating a tank **70**A and a bottle **100**A according to a modification prior to the insertion of the bottle **100**A to the recessed portion **77** of the tank **70**A taken along a cross-section corresponding to the cross-section of FIG. **8**;
(13) FIG. **12** is a cross-sectional view illustrating the tank **70**A and the bottle **100**A inserted in the recessed portion **77** of the tank **70**A taken along the cross-section corresponding to the cross-section of FIG. **8**;
(14) FIG. **13**A is a cross-sectional view of a liquid container **200**;
(15) FIG. **13**B is a cross-sectional view of a liquid container **300**;
(16) FIG. **14**A is a partial cross-sectional view of a tank **70**B including a tank body **71**B and a

recessed portion **77**B protruding upward from an upper wall **76**B of the tank body **71**B; and
(17) FIG. **14**B is a partial cross-sectional view of a tank **70**C including a tank body **71**C and a recessed portion **77**C provided separately from the tank body **71**C.

DESCRIPTION

(18) Hereinafter, one embodiment of the present disclosure will be described with reference to FIGS. **1** through **10**. Incidentally, the embodiment described below is a mere example of the present disclosure, and it would be apparent to those skilled in the art that many modifications and variations may be made thereto without departing from the spirit of the disclosure.
(19) Further, in the following description, upper and lower sides with regard to the multifunction device **10** will be referred to based on an assumption that the multifunction device **10** is disposed on a horizontal surface, i.e., in an operable posture in which the multifunction device **10** is intended to be used (the posture illustrated in FIG. **1**). Specifically, front and rear sides of the multifunction device **10** will be referred to assuming that a surface having an opening **13** of the multifunction device **10** is defined as a front surface of the multifunction device **10** in the operable posture. Left and right sides of the multifunction device **10** will be referred to assuming that the multifunction device **10** in the operable posture is viewed from its front surface.
(20) Further, in the following description, a leftward direction and a rightward direction will be collectively referred to as a left-right direction **9**, wherever appropriate. Likewise, a frontward direction and a rearward direction will be collectively referred to as a front-rear direction **8**, and an upward direction and a downward direction will be collectively referred to as an up-down direction **7**, wherever appropriate. In the present embodiment, the up-down direction **7** is assumed to be vertical, and the front-rear direction **8** and the left-right direction **9** are both assumed to be horizontal. The front-rear direction **8** and the left-right direction **9** are perpendicular to each other.
(21) <Overall Structure of the Multifunction Device **10**>
(22) As illustrated in FIG. **1**, the multifunction device **10** includes a housing **14** having a shape of a generally rectangular parallelepiped. A printer portion **11** is provided in a lower portion of the housing **14**. The multifunction device **10** has various functions such as a facsimile function and a printing function. The multifunction device **10** employs, as the printing function, an inkjet recording system to record an image on one side of a sheet **12**. Incidentally, the multifunction device **10** may be configured to record images on both sides of the sheet **12**.
(23) An operating portion **17** is provided at an upper portion of the housing **14**. The operating portion **17** includes buttons configured to be manipulated to input various commands for recording images and various settings, and a liquid crystal display configured to display various information thereon. In the present embodiment, the operating portion **17** is constituted as a touch panel providing the functions of both the buttons and the liquid crystal display.
(24) As illustrated in FIG. **2**, the printer portion **11** includes a sheet tray **20**, a pick-up unit **16**, an outer guide member **18**, an inner guide member **19**, a pair of conveyer rollers **59**, a pair of discharge rollers **44**, a platen **42**, and a recording unit **24**. These parts of the printer portion **11** are positioned in an interior of the housing **14**. Further, in the interior of the housing **14**, there are provided various condition sensors configured to detect various conditions of the multifunction device **10** and to output signals responsive to the result of the detections. Incidentally, the structure of the printer portion **11** according to the embodiment is merely an example, and may be replaced with other conventional structure.
(25) <Sheet Tray **20**>
(26) As illustrated in FIG. **1**, the opening **13** is open at a front surface **23** of the printer portion **11**. The sheet tray **20** is movable in the front-rear direction **8** through the opening **13** so that the sheet tray **20** can be inserted into and removed from the housing **14**. The sheet tray **20** is movable between a sheet supplying position (illustrated in FIGS. **1** and **2**) where the sheet tray **20** is attached to the housing **14** and a non-sheet supplying position where the sheet tray **20** is pulled out of the housing **14**. More specifically, the sheet tray **20** is inserted rearward relative to the housing **14** to

provide the sheet supplying position, and is withdrawn frontward relative to the housing **14** to provide the non-sheet supplying position.
(27) The sheet tray **20** has a box-like shape whose upper end is open upward, and is configured to accommodate therein the sheets **12**. As illustrated in FIG. **2**, a stack of the sheets **12** is supported on a bottom plate **22** of the sheet tray **20**. A discharge tray **21** is positioned above a front portion of the sheet tray **20**. After an image is recorded on the sheet **12** by the recording unit **24**, the sheet **12** is configured to be discharged onto the discharge tray **21** to be supported on an upper surface of the discharge tray **21**.
(28) As illustrated in FIG. **2**, the sheet **12** supported on the sheet tray **20** is configured to be conveyed to a sheet conveying passage **65** when the sheet tray **20** is at the sheet supplying position.
(29) <Pick-Up Unit **16**>
(30) As illustrated in FIG. **2**, the pick-up unit **16** is positioned below the recording unit **24** and above the bottom plate **22** of the sheet tray **20**. The pick-up unit **16** includes a pick-up roller **25**, a pick-up arm **26**, a power transmission mechanism **27**, and a shaft **28**.
(31) The pick-up roller **25** is rotatably supported by a tip end portion of the pick-up arm **26**. The pick-up arm **26** has a base end portion at which the shaft **28** is provided. The pick-up arm **26** is pivotally movable about an axis of the shaft **28** in directions indicated by a bidirectional arrow **29**. With this structure, the pick-up roller **25** can contact and separate from the sheet tray **20** or an uppermost sheet **12** of the sheet stack supported on the sheet tray **20**.
(32) The power transmission mechanism **27** includes a gear train and is configured to transmit a driving force of a motor (not shown) to the pick-up roller **25** to rotate the same. Accordingly, the uppermost sheet **12** of the sheet stack supported on the bottom plate **22** of the sheet tray **20** at the sheet supplying position (i.e., the uppermost sheet **12** in contact with the pick-up roller **25**) is configured to be conveyed to the sheet conveying passage **65**.
(33) <Sheet Conveying Passage **65**>
(34) As illustrated in FIG. **2**, the sheet conveying passage **65** extends from a rear end portion of the sheet tray **20**. The sheet conveying passage **65** includes a curved portion **33** and a linear portion **34**. The curved portion **33** has a generally U-shape extending diagonally upward and rearward and then making a curve toward the front. The linear portion **34** extends approximately in the front-rear direction **8**.
(35) The curved portion **33** is defined by the outer guide member **18** and the inner guide member **19** facing each other and spaced away from each other by a predetermined interval. The outer guide member **18** and the inner guide member **19** both extend in the left-right direction **9**. The linear portion **34** is partly defined by the recording unit **24** and the platen **42** facing each other in the up-down direction **7** with a predetermined interval therebetween.
(36) The sheet **12** supported by the sheet tray **20** is conveyed along the curved portion **33** by the pick-up roller **25** to reach the pair of conveyer rollers **59** where the sheet **12** is nipped between the pair of conveyer rollers **59**. The nipped sheet **12** is then conveyed frontward toward the recording unit **24** along the linear portion **34**. When the sheet **12** reaches a position immediately below the recording unit **24**, the sheet **12** is subjected to image recordation by the recording unit **24** by deposition of the ink ejected from the recording unit **24** on the sheet **12**. The image-recorded sheet **12** is then conveyed frontward along the linear portion **34** and is discharged onto the discharge tray **21**. In this way, the sheet **12** is conveyed in a conveying direction **15** which is indicated by a one-dotted chain line in FIG. **2**.
(37) <Pair of Conveyer Rollers **59** and Pair of Discharge Rollers **44**>
(38) As illustrated in FIG. **2**, the pair of conveyer rollers **59** is positioned at the linear portion **34**. The pair of discharge rollers **44** is positioned at the linear portion **34** and downstream of the pair of conveyer rollers **59** in the conveying direction **15**.
(39) The pair of conveyer rollers **59** includes a conveyer roller **60** and a pinch roller **61** positioned below the conveyer roller **60**. The pinch roller **61** is urged toward the conveyer roller **60** by a

resiliently urging member such as a coil spring (not illustrated). The pair of conveyer rollers **59** is configured to nip the sheet **12** between the conveyer roller **60** and the pinch roller **61**.
(40) The pair of discharge rollers **44** includes a discharge roller **62** and a spur roller **63** positioned above the discharge roller **62**. The spur roller **63** is urged toward the discharge roller **62** by a resiliently urging member such as a coil spring (not illustrated). The pair of discharge rollers **44** is configured to nip the sheet **12** between the discharge roller **62** and the spur roller **63**.
(41) The conveyer roller **60** and the discharge roller **62** are driven to rotate upon receipt of the driving force from the motor (not shown). As the conveyer roller **60** rotates with the sheet **12** nipped between the conveyer rollers **59**, the sheet **12** is conveyed in the conveying direction **15** onto the platen **42** by the pair of conveyer rollers **59**. As the discharge roller **62** rotates with the sheet **12** nipped by the pair of discharge rollers **44**, the sheet **12** is conveyed in the conveying direction **15** onto the discharge tray **21** by the pair of discharge rollers **44**.
(42) <Platen **42**>
(43) As illustrated in FIG. **2**, the platen **42** is positioned along the linear portion **34** of the sheet conveying passage **65**. The platen **42** faces the recording unit **24** in the up-down direction **7**. The platen **42** is configured to support the sheet **12** conveyed along the sheet conveying passage **65** from below.
(44) <Recording Unit **24**>
(45) As illustrated in FIG. **2**, the recording unit **24** is positioned above the platen **42**. The recording unit **24** includes a carriage **40**, a head **38**, and a tank **70**.
(46) The carriage **40** is movably supported by guide rails **56**, **57**. The guide rails **56**, **57** are spaced away from each other in the front-rear direction **8** and support the carriage **40** such that the carriage **40** is movable reciprocatingly in the left-right direction **9** perpendicular to the conveying direction **15**. The guide rail **56** is positioned upstream of the head **38** in the conveying direction **15**, and the guide rail **57** is positioned downstream of the head **38** in the conveying direction **15**. The guide rails **56**, **57** are supported by a pair of side frames (not illustrated) positioned outward of the linear portion **34** of the sheet conveying passage **65** in the left-right direction **9**. The carriage **40** is movable upon receipt of the driving force from the motor (not shown).
(47) The head **38** is supported by the carriage **40**. The head **38** has a lower surface **68** exposed downward to an outside and facing the platen **42**. The head **38** includes a plurality of nozzles **39**, an ink passage **37**, and piezoelectric elements (not illustrated).
(48) The plurality of nozzles **39** are open at the lower surface **68** of the head **38**. The ink passage **37** connects the plurality of nozzles **39** to the tank **70**. The piezoelectric elements are respectively configured to deform in the ink passage **37** when supplied with power, thereby ejecting ink droplets downward from the corresponding nozzles **39**.
(49) <Tank **70**>
(50) As illustrated in FIG. **2**, the tank **70** is mounted on the carriage **40**. As illustrated in FIGS. **2** through **4**, the tank **70** includes a tank body **71**, a slider **72**, a sealing member **73**, and a cover **74**.
(51) As illustrated in FIG. **2**, the tank **70** is positioned above the head **38**. Incidentally, in the present embodiment, the tank **70** in its entirety is positioned above the head **38**, but the positional relationship between the tank **70** and the head **38** may be suitably modified. In the present embodiment, the recording unit **24** includes the single tank **70**. Black ink is stored in the single tank **70**. Incidentally, the color of the ink stored in the tank **70** need not be black.
(52) As illustrated in FIG. **4**, the tank body **71** has a tank storage chamber **75**. Ink is stored in the tank storage chamber **75**. As illustrated in FIG. **2**, the tank storage chamber **75** of the tank body **71** is in communication with the plurality of nozzles **39** through the ink passage **37**. Accordingly, the ink in the tank storage chamber **75** is supplied to the nozzles **39**. Incidentally, a bottom of the tank body **71** is not depicted in FIG. **4**.
(53) As illustrated in FIGS. **3** and **4**, the tank body **71** includes an upper wall **76** formed with a recessed portion **77** recessed toward the tank storage chamber **75**. The recessed portion **77** has a

hollow cylindrical shape protruding downward from the upper wall **76**, and has a lower open end. That is, the recessed portion **77** is open upward to provide a connection opening **77***a* on the upper wall **76**. A bottle **100** (see FIG. **5**) can be connected to the recessed portion **77** through the opening **77***a* to provide fluid communication between internal spaces of the recessed portion **77** and the bottle **100**. The recessed portion **77** has an outer surface with which the cover **74** is fitted. The cover **74** has a bottom wall formed with an injection opening **78** for injecting the ink to the tank storage chamber **75**. The sealing member **73** is interposed between the cover **74** and the lower end of the recessed portion **77**.

(54) The recessed portion **77** has an upper end portion where three protrusions **79** are positioned, the upper end portion defining the connection opening **77***a* therein. The three protrusions **79** are spaced apart from each other at equal intervals by 120 degrees about an axis **77**X of the recessed portion **77** as a center. Each of the protrusions **79** protrudes toward the axis **77**X from an inner peripheral surface of the upper end portion of the recessed portion **77**.

(55) As illustrated in FIG. **4**, a female thread **80** is formed in an inner peripheral surface of the recessed portion **77**. The female thread **80** is threadingly engageable with a male thread **81** formed on an outer peripheral surface of the slider **72**. As the slider **72** rotates clockwise relative to the recessed portion **77** in a state where the female thread **80** and the male thread **81** are threadingly engaged with each other, the slider **72** moves downward in the internal space of the recessed portion **77**. As the slider **72** rotates counterclockwise relative to the recessed portion **77**, the slider **72** moves upward in the internal space of the recessed portion **77**.

(56) As illustrated in FIG. **4**, the slider **72** includes a slider body **82**, and nozzles **83**, **84**. The slider body **82** has a hollow cylindrical shape and can be accommodated in the recessed portion **77**. The slider body **82** has a lower end closed by a lower end wall **85**. On the outer peripheral surface of the slider body **82**, the male thread **81** is formed. The slider body **82** has a dimension in the up-down direction **7** smaller than a dimension of the internal space of the recessed portion **77** in the up-down direction **7**. As such, the slider body **82** is movable in the up-down direction **7** (upward and downward) in a state where the slider body **82** is accommodated in the internal space of the recessed portion **77**.

(57) The nozzle **83** extends upward from the lower end wall **85** of the slider body **82**, and the nozzle **84** extends downward from the lower end wall **85**. The nozzle **83** has a tip end (upper end) having a sharp triangular shape. The nozzle **84** has a tip end (lower end) constituting a flat surface extending in the front-rear direction **8** and the left-right direction **9**. The lower end of the nozzle **84** is open downward to provide a communication opening **84***a* (see FIGS. **4** and **10**). A partition wall **86** partitions internal spaces of the nozzles **83**, **84** to provide two flow channels **87**, **88** extending in the up-down direction **7**. The flow channel **87** has a constant horizontal cross-sectional area in the up-down direction **7**. With regard to the flow channel **88**, a horizontal cross-sectional area of the internal space of the nozzle **84** is greater than a horizontal cross-sectional area of the internal space of the nozzle **83**. That is, the horizontal cross-sectional area of the flow channel **88** is greater in a lower region below the lower end wall **85** than in an upper region above the lower end wall **85**, such that the horizontal cross-sectional area of the flow channel **88** expands in the lower region to provide a stepwise between the upper and lower regions. Lower openings of the flow channels **87** and **88** constitute the communication opening **84***a* formed in the lower end of the nozzle **84**.

(58) The slider body **82** has an inner peripheral surface where three ribs **89** are positioned. The three ribs **89** are spaced apart from each other at equal intervals by 120 degrees about the axis **77**X of the recessed portion **77** as a center. Each of the ribs **89** protrudes toward the axis **77**X of the recessed portion **77** from the inner peripheral surface of the recessed portion **77**. Each rib **89** extends in the up-down direction **7** to span from the upper end to the lower end of the slider body **82**.

(59) The sealing member **73** is an elastically deformable member made from elastomer or rubber. The sealing member **73** includes a peripheral wall **93** having a flattened hollow cylindrical shape,

as illustrated in FIG. **4**. The sealing member **73** also includes an upper wall **90** provided at an upper end of the peripheral wall **93**, and an abutment plate **91** and a flange **92** provided at a lower end of the peripheral wall **93**.

(60) The upper wall **90** has a disc-like shape having an opening at a center region thereof to enable the nozzle **84** to extends through the opening. Specifically, an inner peripheral surface defining the center opening of the upper wall **90** and an outer peripheral surface of the nozzle **84** are in liquid-tight contact with each other. The abutment plate **91** has an outer diameter smaller than an inner diameter of the peripheral wall **93**. The abutment plate **91** is connected to the peripheral wall **93** through a plurality of connecting portions such that the abutment plate **91** is positioned at a radial center of the peripheral wall **93**. Ink can flow through gaps between the abutment plate **91** and the peripheral wall **93**.

(61) The outer diameter of the abutment plate **91** is slightly greater than an outer diameter of the nozzle **84**. The abutment plate **91** can abut on the tip end (lower end) of the nozzle **84** to close (seal) the tip end of the nozzle **84**. The flange **92** protrudes radially outward from the lower end of the peripheral wall **93**. The flange **92** is nipped between the cover **74** and the recessed portion **77** so as to be immovable in the upward/downward direction. Hence, the internal space of the recessed portion **77** is liquid tightly sealed against the tank storage chamber **75**. The abutment plate **91** is supported by the cover **74** from below. In a state where the tip end of the nozzle **84** is separated from the abutment plate **91** in the up-down direction **7** (as depicted in FIG. **10**), ink and air can flow through a gap between the tip end of the nozzle **84** and the injection opening **78** of the cover **74**.

(62) As illustrated in FIG. **2**, a lid **94** is fitted with the recessed portion **77**. Upon detachment of the lid **94** from the recessed portion **77**, the internal space of the recessed portion **77** is exposed to the outside. With this exposed state, by inserting the bottle **100** in the recessed portion **77** and rotating the bottle **100**, ink is injected from the bottle **100** into the tank storage chamber **75** through the injection opening **78**.

(63) Incidentally, the tank **70** may has an air vent opening (not illustrated). Further, the air vent opening may be opened and closed by an electromagnetic valve, for example.

(64) <Bottle **100**>

(65) Next, the bottle **100** will be described with reference to FIGS. **5** and **6**.

(66) The bottle **100** stores ink therein. The bottle **100** is adapted to supply ink to the tank **70** through the injection opening **78**. A combination of the multifunction device **10** and the bottle **100** is an example of a “liquid supplying system” of the disclosure. As illustrated in FIGS. **5** and **6**, the bottle **100** includes a casing **101**, a sealing member **102**, and a valve **103**.

(67) As illustrated in FIG. **5**, the casing **101** has a generally cylindrical outer shape elongated in the up-down direction **7**. FIGS. **5** and **6** show a posture of the bottle **100** with a supply opening **105** of the bottle **100** facing downward. However, the bottle **100** may be in such a posture that the supply opening **105** faces upward during transportation or storage thereof.

(68) As illustrated in FIG. **6**, the casing **101** has an open lower end. Further, the casing **101** has an upper open end that is sealed by a cap member **111**. The casing **101** has an internal space serving as a storage chamber **104** for storing ink. A spring seat **112** having a solid cylindrical shape is positioned in the storage chamber **104**. The spring seat **112** has an upper surface extending in the front-rear direction **8** and the left-right direction **9**. The spring seat **112** has an upper end portion provided with a flange protruding outward. The spring seat **112** has a lower end surface on which a coil spring **114** is seated. The spring seat **112** has a lower portion inserted in the coil spring **114**. The spring seat **112** has a maximum outer diameter smaller than an inner diameter of the storage chamber **104**. The spring seat **112** is connected to an inner peripheral surface of the casing **101** by a plurality of connection ribs **113**. Ink can flow through gaps between the spring seat **112** and the inner peripheral surface of the casing **101**.

(69) As illustrated in FIG. **6**, the sealing member **102** is positioned inside a lower end portion of the

casing **101**. The sealing member **102** is a hollow cylindrical and elastically deformable member made from rubber or elastomer. The sealing member **102** and the inner peripheral surface of the casing **101** are in contact with each other to provide a liquid tight seal therebetween. The sealing member **102** has upper and lower open ends and defines an internal space serving as the supply opening **105** through with the ink is allowed to flow from the storage chamber **104** to the outside of the storage chamber **104**.
(70) As illustrated in FIG. **6**, the valve **103** is positioned inside the storage chamber **104**. The valve **103** has a generally columnar outer shape. The valve **103** has a lower end portion from which a flange protrudes outward. The flange has an outer diameter smaller than the inner diameter of the storage chamber **104**. Hence, ink can flow along an outer peripheral surface of the valve **103**. The valve **103** is positioned closer to the sealing member **102** than the spring seat **112** is to the sealing member **102**. The coil spring **114** is interposed between the spring seat **112** and the valve **103**. The coil spring **114** has one end in abutment with the flange of the spring seat **112** and another end in abutment with the flange of the valve **103** such that the coil spring **114** is in a compressed state.
(71) Due to an urging force of the coil spring **114**, a lower end surface of the valve **103** is urged to make contact with the sealing member **102** to provide a liquid tight seal to the supply opening **105**. In a case where the nozzle **83** is inserted in the supply opening **105** to come into abutment with the lower end surface of the valve **103** to move the valve **103** away from the sealing member **102** against the urging force of the coil spring **114**, the ink in the storage chamber **104** flows out to the outside through the flow channel **87** of the nozzle **83**.
(72) As illustrated in FIG. **5**, three grooves **115** are formed on the outer peripheral surface of the casing **101**. The three grooves **115** are positioned spaced away from each other at equal intervals by 120 degrees about an axis **100**X of the bottle **100**. Each groove **115** includes a first groove **115**A, and a second groove **115**B. The first groove **115**A extends in the up-down direction **7** and has a lower end that is open at the lower end surface of the casing **101**. The second groove **115**B extends leftward in FIG. **5** in a circumferential direction of the casing **101** from an upper end of the first groove **115**A. The first groove **115**A and the second groove **115**B are in communication with each other to provide a single space in the groove **115**. Each protrusion **79** of the tank **70** can be entered in a corresponding one of the grooves **115**.
(73) <Ink Supply from Bottle **100** to Tank **70**>
(74) Hereinafter, how ink is supplied from the bottle **100** to the tank **70** will be described with reference to FIGS. **7** through **10**.
(75) As the ink in the tank **70** is consumed by ejection of the ink from the nozzles **39** of the head **38**, a user will replenish ink with the tank **70** in response to, for example, an alarm indicating that an amount of the ink in the tank **70** becomes smaller. For replenishing ink with the tank **70**, the user pivotally moves an upper cover of the multifunction device **10** to expose the upper wall **76** of the tank **70** to the outside. The user then removes the lid **94** from the recessed portion **77** to expose the recessed portion **77** to the outside.
(76) The user holds the bottle **100** containing ink so that the supply opening **105** faces downward, and inserts the lower end portion of the bottle **100** into the recessed portion **77** of the tank **70**. At this time, the bottle **100** is in a state where the valve **103** closes the supply opening **105**. Further, in the tank **70**, the slider **72** is at its lowermost position such that the lower end of the nozzle **84** (i.e., lower openings of the flow channels **87**, **88**) is closed by the abutment plate **91** of the sealing member **73**.
(77) As illustrated in FIGS. **7** and **8**, when inserting the lower end portion of the bottle **100** into the recessed portion **77**, the user brings the first grooves **115**A of the casing **101** into alignment with the respective protrusions **79** of the recessed portion **77**. Upon positional alignment of the first groove **115**A with the protrusion **79**, the protrusions **79** are allowed to enter the corresponding first grooves **115**A, and the bottle **100** can be inserted into the recessed portion **77** using the protrusions **79** as a guide.

(78) As illustrated in FIG. **9**, the nozzle **83** of the tank **70** enters the supply opening **105** of the bottle **100** when each of the protrusions **79** reaches the upper end of the corresponding first groove **115**A. The sealing member **102** comes into liquid tight contact with the outer peripheral surface of the nozzle **83**. In this state, the valve **103** closes the supply opening **105**.
(79) Further, as illustrated in FIG. **9**, the ribs **89** of the slider **72** enter the corresponding first grooves **115**A. In this state, the casing **101** is rotatable counterclockwise about the axis **100**X relative to the tank **70** with the guide of the protrusions **79**. As the user rotates the casing **101** counterclockwise, the protrusions **79** enter into the corresponding second grooves **115**B.
(80) In response to the counterclockwise rotation of the casing **101**, the ribs **89** also move counterclockwise because of the engagement between the respective pairs of the first grooves **115**A and the ribs **89**. That is, inside the recessed portion **77**, no idle rotation of the slider **72** relative to the casing **101** occurs, and the slider **72** is ensured to rotate counterclockwise together with the casing **101**. Since the male thread **81** of the slider **72** is threadingly engaged with the female thread **80** of the recessed portion **77**, the slider **72** moves upward within the recessed portion **77** in response to the counterclockwise rotation of the slider **72**, so that the slider **72** reaches its uppermost position as illustrated in FIG. **10**. In accordance with the upward movement of the slider **72**, the ribs **89** move upward relative to the corresponding first grooves **115**A. Since the protrusions **79** are positioned in the corresponding second grooves **115**B during the counterclockwise rotation of the casing **101**, the casing **101** does not move upward relative to the recessed portion **77** in spite of the upward movement of the slider **72**. Further, the casing **101** cannot be pulled out of the recessed portion **77** due to the engagement of the protrusions **79** in the corresponding second grooves **115**B.
(81) As illustrated in FIG. **10**, when the slider **72** reaches its uppermost position inside the recessed portion **77**, the nozzle **83**, which has moved upward together with the slider **72**, pushes the valve **103** upward against the urging force of the coil spring **114**. Accordingly, the valve **103** is separated away from the sealing member **102**, so that the ink in the storage chamber **104** of the bottle **100** can flow into the flow channel **87** of the nozzle **83**. Further, the lower end of the nozzle **84** is also separated away from the abutment plate **91** of the sealing member **73**, thereby allowing ink and air to circulate through the flow channels **87**, **88** of the nozzle **84**. Further, the flow channels **87**, **88** of the nozzle **84** communicate with the injection opening **78** of the cover **74** along a perimeter of the abutment plate **91** of the cover **74**.
(82) As described above, the horizontal cross-sectional area of the flow channel **88** is greater in the lower region below the lower end wall **85** than in the upper region above the lower end wall **85** such that the flow channel **88** forms a stepwise flow channel. Therefore, water head difference is generated between an upper opening of the flow channel **87** and an upper opening of the flow channel **88**. Accordingly, the ink stored in the storage chamber **104** is introduced into the flow channel **87** and flows into the tank storage chamber **75** of the tank **70** through the injection opening **78** of the cover **74**.
(83) As the ink flows into the tank storage chamber **75**, air in the tank storage chamber **75** in turn flows into the storage chamber **104** through the flow channel **88**. Here, a volume of the ink flowing from the storage chamber **104** to the tank storage chamber **75** is approximately equal to a volume of the air flowing from the tank storage chamber **75** to the storage chamber **104**. In this way, so called gas-liquid substitution occurs. The gas-liquid substitution is terminated when all the ink in the storage chamber **104** of the bottle **100** flows into the tank storage chamber **75** of the tank **70**.
(84) Upon termination of the ink supply from the bottle **100** to the tank **70**, the user rotates the casing **101** clockwise relative to the tank **70** from the state illustrated in FIG. **10** to the state illustrated in FIG. **9**. As such, the protrusions **79** of the tank **70** now enter the corresponding first grooves **115**A of the casing **101**, thereby rendering the bottle **100** movable upward relative to the tank **70**. When the bottle **100** is in the state illustrated in FIG. **9**, leakage of ink through the supply opening **105** of the bottle **100** does not occur even if the bottle **100** is detached from the tank **70**,

since the valve **103** closes the supply opening **105**. Further, the slider **72** moves downward within the recessed portion **77**, since the slider **72** rotates clockwise in accordance with the clockwise rotation of the casing **101** relative to the tank **70**. The lower end of the nozzle **84** thus comes into abutment with the abutment plate **91** of the sealing member **73**, thereby closing the flow channels **87**, **88**.

Functions and Advantages of the Embodiment

(85) As the bottle **100** is rotated counterclockwise with the ribs **89** of the slider **72** fitted with the corresponding first grooves **115**A of the bottle **100**, the slider **72** is also rotated and is slidingly moved in the up-down direction **7** together with the nozzles **83**, **84**. When the nozzle **83** of the slider **72** enters into the supply opening **105** of the bottle **100** to open the valve **103**, the ink in the bottle **100** flows into the tank storage chamber **75** of the tank **70** through the nozzles **83**, **84**.

(86) Further, the slider **72** is slidingly moved upward relative to the recessed portion **77** while rotating counterclockwise, by the threading engagement of the slider **72** with the recessed portion **77**. The sealing member **73** can open and close the lower end of the nozzle **84**, depending on the relative position between the sealing member **73** and the slider body **82**.

(87) Further, by the fitting engagement of the protrusions **79** of the tank **70** with the corresponding second grooves **115**B of the bottle **100**, the bottle **100** does not move in the up-down direction **7** relative to the tank **70** during the rotation of the bottle **100** relative to the tank **70**. Hence, the nozzle **83** of the slider **72** can securely enter into the supply opening **105** of the bottle **100**. Further, there is no need for the user to hold the bottle **100** during the entry of the nozzle **83** into the supply opening **105**, in an attempt to prevent the bottle **100** from moving in the up-down direction **7** against the urging force of the coil spring **114**. Further, pop up of the bottle **100** relative to the tank **70** due to the urging force of the spring **114** is less likely to occur for detachment of the bottle **100** from the tank **70**.

Modifications to the Embodiment

(88) According to the above-described embodiment, the sealing between the slider **72** and the recessed portion **77** is provided by the liquid tight contact between the nozzle **84** of the slider **72** and the sealing member **73**. However, the liquid tight sealing between the slider **72** and the recessed portion **77** may be provided by a different configuration from the embodiment.

(89) As an example, FIGS. **11** and **12** depict a tank **70**A and a bottle **100**A according to a modification to the embodiment. In this modification, the tank **70**A includes a slider **72**A and a sealing member **95**, instead of the slider **72** and the sealing member **73** of the embodiment. Unlike the slider **72**, the slider **72**A according to this modification does not include the nozzle **84**, and includes a nozzle **83**A in place of the nozzle **83** of the embodiment. The sealing member **95** is provided in the recessed portion **77** of the tank **70**A, in place of the sealing member **73** of the embodiment.

(90) The sealing member **95** is an elastically deformable member made from elastomer or rubber. The sealing member **95** includes a peripheral wall **96** having a generally flattened conical shape, as illustrated in FIGS. **11** and **12**. The peripheral wall **96** has an upper open end. While making elastic deformation, the peripheral wall **96** is in contact with the lower end wall **85** in a region around the lower end opening of the nozzle **83**A of the slider **72**A. The sealing member **95** also includes an abutment portion **97** positioned at an internal space surrounded by the peripheral wall **96**. The abutment portion **97** has a generally solid tapered shape. The abutment portion **97** closes the lower end opening of the nozzle **83**A when the slider **72**A is at its lowermost position (illustrated in FIG. **11**). The abutment portion **97** has an outer diameter smaller than an inner diameter of the lower end portion of the peripheral wall **96**. The abutment portion **97** is connected to the peripheral wall **96** through a plurality of connecting portions such that the abutment portion **97** is positioned at a diametrical center of the peripheral wall **96**. Ink can flow through gaps between the abutment portion **97** and the peripheral wall **96**. A flange **98** protrudes outward from the lower end portion of the peripheral wall **96**. The flange **98** is nipped between the cover **74** and the recessed portion **77** so

as to be immovably fixed in the up-down direction **7**. In this way, the internal space of the recessed portion **77** is liquid-tightly sealed against the tank storage chamber **75**.

(91) As illustrated in FIG. **12**, the lower end opening of the nozzle **83**A is separated from the abutment portion **97** when the slider **72**A is at its uppermost position. At this time, the upper end of the peripheral wall **96** is in contact with the lower end wall **85** of the slider **72**A due to the elastic restoration of the peripheral wall **96**. As such, ink can flow from the lower end opening of the nozzle **83**A to the injection opening **78** of the cover **74**.

(92) As illustrated in FIGS. **11** and **12**, unlike the nozzle **83** of the above-described embodiment, the nozzle **83**A of the slider **72**A may not be partitioned into two channels but defines a single channel therein.

(93) The bottle **100**A includes a casing **101**A, and a cap member **111**A in place of the cap member **111** of the embodiment. The cap member **111**A is formed with an air communication passage **106**. The air communication passage **106** may be sealed with a semi-permeable membrane, or may has a labyrinth structure, or may be opened and closed by a valve in order to avoid leakage of ink. As illustrated in FIGS. **11** and **12**, the coil spring **114** may be positioned in the storage chamber **104** of the bottle **100**A such that the coil spring **114** is interposed between a valve **103**A and the cap member **111**A. In this case, a shape of the valve **103**A may be suitably modified, provided that the valve **103**A can open and close the supply opening **105** of the bottle **100**A.

Other Variations and Modifications

(94) (1) Incidentally, according to the above-described embodiment, the casing **101** of the bottle **100** (liquid container) is hollow cylindrical, but the shape of the liquid container need not be limited to that of the embodiment. For example, the casing **101** may have a bottle-like shape other than the hollow cylindrical shape. For example, FIGS. **13**A and **13**B respectively depict liquid containers whose shapes are different from the hollow cylindrical shape of the bottle **100** of the embodiment.

(95) Referring to FIG. **13**A, a liquid container **200** includes a container **201** having a narrow mouth and produced by blow molding, and a supply portion **202** threadingly engaged with the narrow mouth. The supply portion **202** includes a sealing member **203**, a valve **204**, and a coil spring **205** corresponding to the sealing member **102**, the valve **103**, and the coil spring **114** of the embodiment, respectively. The liquid container **200** is configured to supply liquid to the tank **70** provided with the nozzle **83** having two flow channels **87**, **88**.

(96) Referring to FIG. **13**B, a liquid container **300** includes a pouch **301** produced by welding two resin sheets together, and a supply portion **302** connected to the pouch **301** by welding. The supply portion **302** includes a sealing member **303**, a valve **304**, and a coil spring **305** corresponding to the sealing member **102**, the valve **103**, and the coil spring **114** of the embodiment, respectively. The liquid container **300** according to this variation is configured to supply liquid to the tank **70**A including the nozzle **83**A without the partition wall **86** providing a single flow channel (see FIGS. **11** and **12**).

(97) (2) With respect to the tank **70** of the embodiment, the recessed portion **77** is formed at the upper wall **76**. However, the recessed portion of the disclosure may be formed at a portion other than the upper wall of the tank. For example, the recessed portion may be formed at an outer surface of the tank and at a sloped wall inclined with respect to the up-down direction **7**. Further, the recessed portion may not be recessed from the upper wall. For example, FIG. **14**A illustrates a recessed portion **77**B according to a variation of the recessed portion **77** of the embodiment. The recessed portion **77**B is a hollow cylindrical portion protruding upward from an upper wall **76**B of a tank body **71**B of a tank **70**B. Still further, the tank of the disclosure need not be mounted on the carriage **40**. Instead, the tank may be connected to the head **38** to provide fluid communication of ink therebetween by a tube, for example.

(98) (3) Further, each of the tank body **71**, the slider **72**, and the cover **74** may not be an integral product, but a plurality of members may be assembled together to constitute the same. For

example, as illustrated in FIG. **14**B, a tank **70**C includes a tank body **71**C and a recessed portion **77**C respectively constituted by separate members. Similarly, the casing **101** and the valve **103** may not be integral products, but may be constituted by assemblies of separate members.
(99) (4) Further, in the above-described embodiment, ink is used as one example of liquid for printing. However, the liquid for printing is not limited to the ink. For example, the liquid for printing may be: pretreatment liquid to be ejected to printing sheets prior to ink ejection; and water to be sprayed for avoiding dewatering of the nozzle **39** of the head **38**.
(100) While the description has been made in detail with reference to the embodiments, it would be apparent to those skilled in the art that many modifications and variations may be made thereto.
Remarks
(101) The multifunction device **10** is an example of a liquid supplying device. The tanks **70**, **70**A, **70**B, **70**C are examples of a tank. The tank bodies **71**, **71**B, **71**C are examples of a tank body. The tank storage chamber **75** is an example of a tank storage chamber. The recessed portions **77**, **77**B, **77**C are examples of a recessed portion. The connection opening **77***a* of the recessed portions **77**, **77**B, **77**C is an example of a connection opening. The sliders **72**, **72**A are examples of a slider. The nozzles **83**, **84** are examples of a nozzle. The communication opening **84***a* is an example of a communication opening. The sealing members **73**, **95** are examples of a sealing member. The injection opening **78** is as an example of an injection opening. The rib **89** is an example of a fitting portion and an example of a first fitting portion. The protrusion **79** is as an example of a second fitting portion. The bottle **100** and the liquid containers **200**, **300** are examples of a liquid container. The storage chamber **104** is an example of a container storage chamber. The coil spring **114** is an example of a resiliently urging member. The groove **115** is an example of a counter-fitting portion. The first groove **115**A is an example of a first part, and the groove **115**B is an example of a second part of the counter-fitting portion. The lowermost position of the slider **72** is an example of a first position, and the uppermost position of the slider **72** is an example of a second position.

Claims

1. A tank to which a liquid container storing liquid to be supplied to the tank is connectable, the tank comprising: a tank body defining a tank storage chamber configured to store the liquid therein; and a slider including a nozzle connectable to the liquid container and extending in an extending direction, the nozzle having a communication opening configured to communicate with the tank storage chamber of the tank body, the slider being rotatable about an axis extending in the extending direction relative to the tank body and being movable in the extending direction relative to the tank body to open and close the communication opening in accordance with rotations of the slider.
2. The tank according to claim 1, further comprising: a fitting portion provided at the slider and configured to be fitted with the liquid container.
3. The tank according to claim 1, further comprising: a recessed portion provided at the tank body and having an injection opening in communication with the tank storage chamber, the recessed portion having a connection opening through which the liquid container is connectable, wherein the slider is positioned in the recessed portion and is rotatable relative to the recessed portion, and the communication opening of the nozzle is configured to communicate with the injection opening of the recessed portion; and a sealing member configured to provide a liquid tight sealing to a gap between the slider and the recessed portion, and configured to also allow the liquid to flow between the communication opening of the nozzle and the injection opening of the recessed portion.
4. The tank according to claim 3, wherein the slider comprises a slider body threadingly engaged with the recessed portion, and the slider being movable between a first position and a second position, the second position being closer to the connection opening than the first position is to the connection opening in the extending direction, wherein the sealing member is configured to close

the communication opening of the nozzle when the slider body is at the first position, and wherein the sealing member is configured to open the communication opening of the nozzle when the slider body is at the second position.
5. The tank according to claim 1, wherein the nozzle defines therein a first channel and a second channel.
6. The tank according to claim 5, wherein the first channel has a horizontal cross-sectional area that is constant in the extending direction, and wherein the second channel comprises: a first portion whose horizontal cross-sectional area is constant in the extending direction; and a second portion whose horizontal cross-sectional area is greater than the horizontal cross-sectional area of the first portion, the second portion being closer to the communication opening than the first portion is to the communication opening in the extending direction.
7. A liquid supplying system comprising a liquid container and a tank, wherein the liquid container comprises: a casing; a container storage chamber defined in the casing and storing therein liquid to be supplied to the tank; a supply opening providing fluid communication between the container storage chamber and an outside of the casing; and a counter-fitting portion provided at the casing, wherein the tank comprises: a tank body defining a tank storage chamber configured to store liquid therein the liquid supplied from the container storage chamber; and a slider including a nozzle connectable to the liquid container and extending in an extending direction, the nozzle having a communication opening configured to communicate with the tank storage chamber of the tank body, the slider being rotatable about an axis extending in the extending direction relative to the tank body and being movable in the extending direction relative to the tank body to open and close the communication opening in accordance with rotations of the slider.
8. The liquid supplying system according to claim 7, wherein the liquid container further comprises: a valve configured to open and close the supply opening; and a resiliently urging member urging the valve toward the supply opening.
9. The liquid supplying system according to claim 7, wherein the casing has an air communication passage configured to provide communication between the container storage chamber and the outside of the casing.
10. The liquid supplying system according to claim 7, further comprising: a first fitting portion provided at the slider and configured to be fitted with the liquid container.
11. The liquid supplying system according to claim 10, further comprising: a recessed portion provided at the tank body and having an injection opening in communication with the tank storage chamber, the recessed portion having a connection opening through which the liquid container is connectable, wherein the slider is positioned in the recessed portion and is rotatable relative to the recessed portion, and the communication opening of the nozzle is configured to communicate with the injection opening of the recessed portion; and a sealing member configured to provide a liquid tight sealing to a gap between the slider and the recessed portion, and configured to also allow the liquid to flow between the communication opening of the nozzle and the injection opening of the recessed portion.
12. The liquid supplying system according to claim 11, wherein the slider comprises a slider body threadingly engaged with the recessed portion, and the slider being movable between a first position and a second position, the second position being closer to the connection opening than the first position is to the connection opening in the extending direction, wherein the sealing member is configured to close the communication opening of the nozzle when the slider body is at the first position, and wherein the sealing member is configured to open the communication opening of the nozzle when the slider body is at the second position.
13. The liquid supplying system according to claim 10, wherein the counter-fitting portion comprises: a first part extending in the extending direction and open at a lower end of the casing; and a second part extending from the first part in a direction following the rotations of the slider, wherein the tank further comprises a second fitting portion provided at the recessed portion and

configured to be fitted with the counter-fitting portion of the liquid container, wherein the first fitting portion is configured to be fitted with the first part to restrict the liquid container from rotating relative to the slider, and wherein the second fitting portion is configured to be fitted with the second part to allow the liquid container to rotate relative to the recessed portion but to restrict the liquid container from moving in the extending direction relative to the recessed portion.

14. The liquid supplying system according to claim 13, wherein the first fitting portion is a rib provided on an inner surface of the slider body, and wherein the second fitting portion is a protrusion provided around the connection opening of the recessed portion to protrude from an inner surface of the recessed portion.

15. The liquid supplying system according to claim 7, wherein the nozzle defines therein a first channel and a second channel.

16. The liquid supplying system according to claim 15, wherein the first channel has a horizontal cross-sectional area that is constant in the extending direction, and wherein the second channel comprises: a first portion whose horizontal cross-sectional area is constant in the extending direction; and a second portion whose horizontal cross-sectional area is greater than the horizontal cross-sectional area of the first portion, the second portion being closer to the communication opening than the first portion is to the communication opening in the extending direction.